C-HI 5 Myr Cavity-Protocol Stub

Date: 2026-05-10

Summary

This report records the first explicit cavity-protocol smoke product for the particles-enabled high-power cluster case, `C-HI`, using the same target as the truth and observable smokes:

- grid output: `data.0008.flt.h5`
- grid timestep: `Timestep_8`
- physical time: `4.9994 Myr`
- view: `0 deg`, projected along the simulation `y` axis
- tracer threshold: `5e-3`

The extraction was run on a `kunanyi` compute node as PBS job `690620.kunanyi-ohpc` and completed successfully. It is a cavity-protocol stub, not the final science cavity measurement: it establishes a reproducible ellipsoid-plus-ambient-shell path against the already aligned 5 Myr target.

Method

The extractor fits one 3D tracer-defined ellipsoid per lobe. For each side of the bipolar source, it uses all cells with `tracer > 5e-3` on that side of the mid-plane to define the lobe centre and semi-axes. It then estimates the local ambient medium from a same-side ellipsoidal shell outside the fitted lobe:

- shell ellipsoidal radius: `1.00` to `1.35`
- inner shell band: `1.00` to `1.15`
- outer shell band: `1.15` to `1.35`
- ambient tracer exclusion: `tracer <= 1e-3`

For each lobe the extractor reports:

- projected ellipsoid boundary for the `0 deg` view
- ellipsoid volume and tracer-extent volume
- local ambient pressure and density
- `pV` and `4pV`
- sound-crossing age
- buoyancy and refill age estimates

The buoyancy/refill ages use a provisional hydrostatic pressure-gradient proxy for gravity from the inner and outer ambient-shell bands. These are useful only as smoke-test values until the final cavity protocol uses the agreed environment/gravity treatment.

Kunanyi Job

- job id: `690620.kunanyi-ohpc`
- job name: `c_hi_cavity5`
- node: `n001`
- resources: `select=1:ncpus=1:mpiprocs=1`
- walltime requested: `00:30:00`
- extraction window: `2026-05-10T08:43:49Z` to `2026-05-10T08:44:25Z`
- extraction result: no stderr output, archive written successfully

Compact archive:

```
remote: /u/bwbrook/codex_kunanyi/c_hi_cavity_stub_2026-05-10.tar.gz
local:  build/science/kunanyi_stage_2026-05-08/c_hi_cavity_stub_690620_compact_2026-05-10.tar
sha256: 8166ab35bf27f7058bb5b44a387fd0f169edc8ccf2e4bbab920228bf3d78f769
```

Scalar Smoke Metrics

Metric	`-z` lobe	`+z` lobe
lobe centre `z`	`-16.535 kpc`	`16.535 kpc`
semi-axis `x`	`9.465 kpc`	`9.465 kpc`
semi-axis `y`	`9.465 kpc`	`9.465 kpc`
semi-axis `z`	`16.510 kpc`	`16.510 kpc`
ellipsoid volume	`6194.84 kpc^3`	`6194.84 kpc^3`
tracer-extent volume	`5395.89 kpc^3`	`5395.89 kpc^3`
ambient pressure	`1.7422e-10 dyn cm^-2`	`1.7422e-10 dyn cm^-2`
ambient density	`2.0091e-26 g cm^-3`	`2.0091e-26 g cm^-3`
`pV`	`3.1709e58 erg`	`3.1709e58 erg`
`4pV`	`1.2684e59 erg`	`1.2684e59 erg`
sound speed	`1202.2 km s^-1`	`1202.2 km s^-1`
sound age	`13.45 Myr`	`13.45 Myr`
buoyancy age	`3.93 Myr`	`3.93 Myr`
refill age	`12.29 Myr`	`12.29 Myr`

The near-identical values are expected for the current centred symmetric `C-HI` control and are a useful sanity check on the measurement path.

Figure

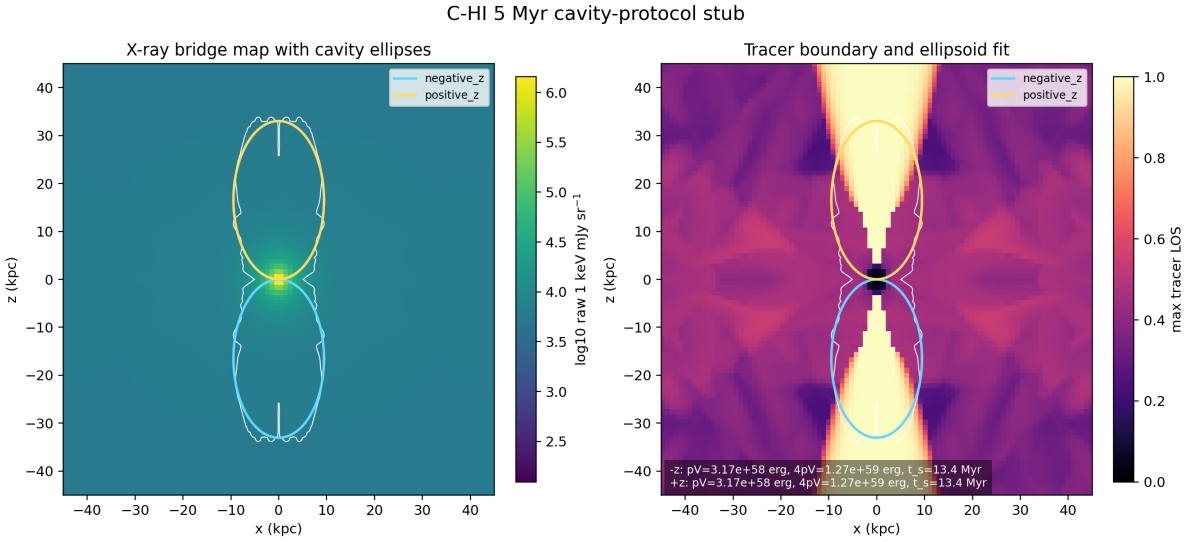


Figure 1: C-HI cavity-protocol stub

The white contour marks the line-of-sight maximum tracer boundary at `tracer = 5e-3`. The coloured ellipses are the fitted projected cavity boundaries derived from the 3D tracer extents.

Products

The compact products are stored under `docs/science/assets/c_hi_cavity_stub_2026-05-10/`:

- `cavity_stub_summary.csv`
- `cavity_stub_metadata.txt`
- `cavity_ellipse_boundary_0deg.tsv`
- `c_hi_cavity_stub_overlay.png`

The extractor and plotting helper are:

- `scripts/science/pluto_h5_cavity_stub_extract.c`
- `scripts/science/plot_cavity_stub.py`
- `scripts/kunanyi/c_hi_cavity_stub.pbs`

Interpretation

This completes the first cavity-protocol smoke gate for the aligned 5 Myr target:

- the same `C-HI` snapshot/view now has Level 1 truth, Level 2 tracer truth, bridge X-ray/radio observable products, and a first cavity measurement stub
- the fitted cavity geometry is symmetric and well inside the current domain
- the local ambient-shell estimator returns finite, symmetric pressure and density values
- `pV`, `4pV`, and simple age estimates can be produced reproducibly on a compute node without moving the full HDF5 file to the MacBook

The main remaining limitation is physics completeness, not mechanics. The buoyancy/refill ages need replacement or calibration against the final agreed environment/gravity treatment, and the bridge X-ray/radio products still need the full GRAiLE/XRAiSE runtime rather than the compact C smoke path.